Supplementary Pilot Summary: Parts 1B, 2B and 3B

Health Demography Portfolio — NIDI-KNAW Health Demography Application

This supplementary report adds three important extensions to the main pilot portfolio: an age-stratified Dutch municipal analysis, a sex-stratified European education–obesity appendix, and a capital-city green infrastructure comparison. Together, these additions strengthen the demographic and environmental relevance of the pilot by adding age, sex and urban green-environment dimensions.

Part 1B: Age-Stratified Dutch Municipal Analysis

Pseudo-life-course analysis using CBS/RIVM Gezondheidsmonitor 2024

Aim

Part 1B asks whether Dutch municipalities with higher adult obesity also show stronger age-related accumulation of long-term conditions and lower physical activity across age groups. Because the data are cross-sectional, this is interpreted as a pseudo-life-course or age-profile analysis, not a true longitudinal trajectory.

Data and sample

The analysis uses CBS/RIVM Gezondheidsmonitor 2024 data for 37 Dutch municipalities with complete information for:

Age group	Description
18–65 years	Working-age adults
65+ years	Older adults
Total adults	Overall municipal adult population

Main indicators:

Indicator	Meaning
Obesity prevalence	Adults with severe overweight/obesity
Physical activity adherence	Adults meeting physical activity guidelines

Indicator	Meaning
Long-term conditions	Adults with one or more long-term conditions
Good/very good health	Self-rated health

Key quantitative findings

Comparison	n	r	p-value	Interpretation
Total adult obesity vs 65+ obesity	37	+0.655	0.000011	Strong positive association
Total adult obesity vs 65+ physical activity	37	-0.626	0.000034	Strong negative association
Total adult obesity vs 65+ long-term conditions	37	+0.043	0.799	No meaningful association
Total adult obesity vs 65+ good/very good health	37	-0.322	0.052	Weak-to-moderate negative association
Total adult obesity vs 65+ minus 18–65 obesity gap	37	-0.483	0.0025	Higher-obesity municipalities have smaller age gaps
Total adult obesity vs 65+ minus 18–65 LTC gap	37	+0.204	0.225	Weak and unclear
Total adult obesity vs 65+ minus 18–65 physical activity gap	37	+0.148	0.381	Weak and unclear

Age-profile pattern by municipal obesity group

Municipality group	18–65 obesity %	65+ obesity %	Obesity age gap	18–65 PA %	65+ PA %	PA age gap	18–65 LTC %	65+ LTC %	LTC age gap
Lower-obesity municipalities	10.98	14.18	+3.20	55.45	45.17	-10.28	28.73	47.85	+19.13
Middle-obesity municipalities	13.48	14.99	+1.52	53.28	41.28	-11.99	29.05	48.01	+18.96
Higher-obesity municipalities	16.92	16.98	+0.07	47.79	39.01	-8.78	28.12	48.05	+19.93

Interpretation

The strongest finding is not that high-obesity municipalities show steeper long-term-condition accumulation with age. Long-term conditions rise sharply from 18–65 to 65+ in all municipality groups, by approximately 19 percentage points. Instead, the key finding is that higher-obesity municipalities already have elevated obesity before age 65, and this disadvantage persists into older adulthood. They also show consistently lower physical activity in both age groups.

This suggests a persistent adult-age municipal disadvantage rather than an obesity pattern that emerges only in later life. The analysis strengthens the life-course relevance of the pilot, while also showing why true longitudinal data such as SHARE or Lifelines are needed to test individual disease trajectories.

Part 2B: Sex-Stratified Educational Inequalities in Obesity

Demographic sensitivity extension of the Eurostat/EHIS comparison

Aim

Part 2B extends the Eurostat/EHIS 2019 comparison by stratifying education–obesity gradients by sex. This is important because sex is a core demographic dimension of health inequality, and total-population education gradients can hide different patterns among women and men.

Data and scope

The analysis is based on Eurostat/EHIS 2019 BMI data for adults aged 18+ in:

Country	Welfare-regime role
Netherlands	Main host-country / corporatist comparator
Finland	Nordic comparator
Poland	Post-socialist comparator

The appendix compares obesity prevalence by:

Dimension	Categories
Sex	Women, men, total
Education	Low, medium, high
Outcome	Obesity, BMI ≥ 30

Table structure for final Part 2B

Country	Sex	Low education obesity %	Medium education obesity %	High education obesity %	Low-high gap	Low/ high ratio
Netherlands	Women	[insert]	[insert]	[insert]	[insert]	[insert]
Netherlands	Men	[insert]	[insert]	[insert]	[insert]	[insert]
Finland	Women	[insert]	[insert]	[insert]	[insert]	[insert]
Finland	Men	[insert]	[insert]	[insert]	[insert]	[insert]
Poland	Women	[insert]	[insert]	[insert]	[insert]	[insert]
Poland	Men	[insert]	[insert]	[insert]	[insert]	[insert]

Interpretation template

Part 2B adds a sex-stratified demographic sensitivity check to the education–obesity comparison. Instead of treating the education gradient as uniform across the adult population, it asks whether socio-economic inequalities in obesity differ between women and men across the Netherlands, Finland and Poland. This is important for a health-demography interpretation because education, body weight, health behaviour and chronic disease risk may be gendered differently across welfare regimes.

If the sex-stratified gradients are steeper among women, this would suggest that education may operate more strongly through gendered pathways such as labour-market position, family responsibility, health behaviour, care burden and access to preventive health resources. If gradients are similar for men and women, the result would support a broader socio-economic pattern that is less sex-specific. In either case, the appendix strengthens the demographic quality of the pilot by preventing total-population estimates from masking sex-specific inequality patterns.

Methodological note

A full doctoral analysis should extend this section using Slope Index of Inequality and Relative Index of Inequality, ideally with individual-level EHIS, SHARE or GGP microdata. In this pilot, the sex-stratified appendix is used as a descriptive demographic sensitivity check.

Part 3B: Capital-City Green Infrastructure and Urban Tree Cover

Environmental-health pathway extension using European Environment Agency data

Aim

Part 3B strengthens the environmental section by adding a more direct health-pathway indicator: urban green infrastructure. The earlier Part 3 indicators — PM2.5-related premature deaths and renewable-energy share — captured air-pollution burden and structural green-transition progress. Part 3B adds urban green infrastructure and tree cover, which are more closely connected to physical activity, heat mitigation, walkability, restoration and environmental exposure.

Data source and cities

The analysis uses European Environment Agency capital-city green infrastructure indicators. For the three-country comparison, the relevant capitals are:

Country	Capital city
Netherlands	Amsterdam
Finland	Helsinki
Poland	Warsaw

Key results

Country	Capital city	Total green infrastructure %	Urban green space %	Urban tree cover %
Finland	Helsinki	62	5	44
Poland	Warsaw	47	6	40
Netherlands	Amsterdam	31	5	14

Combined environmental context

Country	PM2.5 deaths per 100k	Renewable energy share %	Capital green infrastructure %	Urban tree cover %
Finland	0.6	52.1	62	44
Netherlands	21.5	20.2	31	14
Poland	67.2	17.8	47	40

Interpretation

Part 3B shows that environmental-health context is multidimensional. Helsinki has the strongest overall profile: very low PM2.5-related premature mortality, the highest renewable-energy share, the highest capital-city green infrastructure and the highest urban tree cover. Amsterdam has lower green infrastructure and tree cover than both Helsinki and Warsaw, despite the Netherlands having a lower PM2.5 burden than Poland. Warsaw shows a mixed profile: relatively high capital-city green infrastructure and tree cover, but very high PM2.5-related health burden and low renewable-energy share.

This suggests that green-transition context cannot be captured by renewable-energy share alone. Air quality, energy transition and urban green infrastructure represent different environmental-health pathways. For the SEHealth/SoGreen research agenda, this is important because the relevant question is not whether “green transition” directly lowers obesity or cardiometabolic risk, but through which specific pathways — air pollution, green space, walkability, temperature exposure, physical activity infrastructure and food environments — environmental change shapes health inequalities.

Caution

The capital-city green infrastructure data should be interpreted as contextual and hypothesis-generating. Amsterdam, Helsinki and Warsaw are not national averages. They provide comparable urban environmental indicators, but they should not be treated as direct national causal predictors of obesity or chronic disease.

Integrated contribution of Parts 1B, 2B and 3B

Together, these three additions improve the demographic and environmental depth of the pilot.

Part 1B adds an age-profile perspective, showing that high-obesity municipalities have persistent adult-age disadvantage rather than only late-life accumulation. Part 2B adds sex-stratification, strengthening the demographic interpretation of education–obesity gradients across welfare contexts. Part 3B adds a direct urban environmental pathway through green infrastructure and tree cover, improving the earlier environmental section beyond PM2.5 and renewable energy alone.

The combined message is that cardiometabolic health inequalities should be studied across age, sex, socio-economic position, welfare context and environmental exposure. This directly supports a future longitudinal SEHealth/SoGreen-style analysis using SHARE, GGP, ESS or Lifelines linked with geospatial indicators of pollution, green space, climate exposure and built environment.